E-MARKET MANAGEMENT SYSTEM

FINALIZE MVP SCOPE & USER STORIES

SCRUM-10 | Scope Baseline for MVP Development

Project	E-Market Management System
Project Code	EMM-2026-MVP
Duration	06 tuần
Purpose	Chốt phạm vi MVP, user stories, acceptance criteria và business rules để làm baseline cho Sprint 2-6.

Core MVP Flow

Browse Product → Register/Login → Cart → Checkout → Create Order → Manage Order Status

1. Description

Finalize the MVP scope of E-Market Management System, confirm which features are included/excluded, define the main user stories, acceptance criteria, business rules, and priorities so that the development team can implement the next sprints without ambiguity.

The MVP must cover the full core e-commerce flow and clearly separate MUST, SHOULD, COULD, and Post-MVP features. The result of this task is the scope baseline for Sprint 2-6.

2. In Scope - MVP

Authentication

- User registration
- User login
- JWT authentication
- USER / ADMIN authorization

Product

- View product list
- View product details
- Create / Update / Delete product
- Link product with Category and Brand
- Filter products by Category / Brand

Cart

- View cart
- Add item
- Update quantity
- Remove item
- Validate product stock
- Use authenticated user from JWT instead of hard-coded userId

Order

- Checkout
- Create Order and OrderItems
- Calculate total amount on backend
- Validate stock before order creation
- Deduct inventory
- Clear cart after successful checkout
- View order history
- Admin updates order status

Frontend

- Login/Register UI
- Product listing
- Cart UI
- Checkout form
- Toast notifications
- Loading and error states

QA

- Unit testing

- Integration testing
- Responsive testing
- UAT

3. Out of Scope - Post MVP

- Real payment gateway integration (MoMo, VNPAY, Credit Card)
- Real logistics / shipping provider integration
- Advanced Admin Analytics Dashboard
- Recommendation Engine
- Complex Coupon System
- Loyalty / Membership Point
- Advanced Media Management
- Microservices Architecture
- Review & Rating nếu MVP chưa ổn định

4. User Stories & Acceptance Criteria

US-01 - Register Account	MUST

As a Guest
I want to register an account
So that use customer features of the system.

Acceptance Criteria

- User can submit valid registration information.
- Email must be unique.
- Password must be hashed using BCrypt.
- Invalid input returns a clear validation error.
- Successful registration creates a new user account.
- The account receives the default USER role.
- Password/hash is never returned in API responses.
- Authentication token is returned if auto-login is supported.

US-02 - Login	MUST

As a Registered user
I want to log in
So that access cart, orders, and protected functions.

Acceptance Criteria

- Valid credentials return a JWT token.
- Invalid credentials return HTTP 401.
- JWT contains sufficient information to identify/authenticate the user.
- Protected endpoints reject requests without valid authentication.
- Password/hash is never included in API responses.

US-03 - View Product List	MUST

As a Customer
I want to view available products
So that browse products before purchasing.

Acceptance Criteria

- Product list is retrieved from PostgreSQL.
- Product information and price are displayed correctly.
- Product stock information is available to the backend.
- Inactive/unavailable products are handled according to system rules.
- No products returns a valid empty result, not an application error.

US-04 - Add Product to Cart	MUST

As a Logged-in customer
I want to add a product to my cart
So that purchase it later.

Acceptance Criteria

- User must be authenticated.
- Product must exist.
- Quantity must be greater than 0.
- Requested quantity must not exceed available stock.
- Cart belongs to the currently authenticated user.
- User ID comes from JWT/SecurityContext and is never hard-coded.
- Adding the same product updates quantity instead of creating unnecessary duplicates.
- Invalid stock returns a clear business error.

US-05 - Update / Remove Cart Item	MUST

As a Logged-in customer
I want to update or remove cart items
So that control what I want to purchase.

Acceptance Criteria

- User can update quantity.
- Quantity remains valid and does not exceed stock.
- User can remove an item from cart.
- Changes are persisted immediately.
- User can modify only their own cart.
- Unauthorized access to another user's cart is rejected.

US-06 - Checkout / Create Order	MUST

As a Logged-in customer
I want to checkout my cart
So that create an order.

Acceptance Criteria

- User must be authenticated.
- Cart must not be empty.
- Shipping information must be valid.
- Backend loads the current user's cart.
- Backend validates stock again before order creation.
- Backend calculates final total; frontend is not trusted to determine it.
- Order and OrderItem records are created.
- Each OrderItem stores purchase-time price information.
- Product stock is deducted.
- Cart is cleared only after successful checkout.
- If any checkout step fails, the transaction is rolled back.
- Successful checkout returns an Order response.

US-07 - Admin Update Order Status	MUST

As a Admin
I want to update order status
So that manage the order lifecycle.

Acceptance Criteria

- Only ADMIN can update order status.
- Invalid status transitions are rejected.
- MVP status flow includes PENDING → PROCESSING → COMPLETED.
- CANCELLED is supported only if cancellation/stock-return rules are defined.
- Customer users cannot update order status.
- Updated status is persisted correctly.

US-08 - Filter Products by Category / Brand	SHOULD

As a Customer
I want to filter products by Category or Brand
So that find relevant products more easily.

Acceptance Criteria

- Category filter returns matching products.
- Brand filter returns matching products.
- Combined filters return correct results if supported.
- No matches returns a valid empty list.
- Invalid filter parameters are handled correctly.

US-09 - View Order History	SHOULD

As a Logged-in customer
I want to view my previous orders
So that track purchase history.

Acceptance Criteria

- User must be authenticated.
- User sees only orders created by their own account.
- User cannot access another user's orders.
- Order list displays at least Order ID, Created Date, Total Amount, and Status.
- Order detail can display purchased items.
- Unauthorized access is rejected.

US-10 - Review & Rating	COULD / POST-MVP

As a Customer who purchased a product
I want to rate and review the product
So that share feedback.

Acceptance Criteria

- Only eligible users who purchased the product can review it.
- Rating value must be valid.
- Review content follows validation rules.
- Review is linked to both user and product.

5. Business Rules

- Authenticated user identity must come from JWT/SecurityContext.
- Client must not be allowed to decide its own userId.
- Order total must be calculated by backend.
- Product stock must never become negative.
- Cart quantity must be greater than 0 and cannot exceed stock.
- OrderItem must preserve the purchase-time unit price.
- Customer can access only their own cart/orders.
- Admin-only endpoints must verify ADMIN role.
- Checkout must execute inside a database transaction.
- Failure during checkout must rollback the entire transaction.
- API errors must use consistent HTTP status and error response format.

6. Priority Classification

Priority	Features
MUST	Register; Login; JWT Authentication; Product Listing; Cart Operations; Checkout; Order Creation; Stock Deduction; Admin Order Status
SHOULD	Category/Brand Filter; Order History
COULD	Review & Rating
NOT IN MVP	Real Payment; Logistics Integration; Recommendation Engine; Complex Coupon; Loyalty; Microservices; Advanced Dashboard

7. Definition of Done

☑ MVP in-scope features are documented.
☑ Out-of-scope features are documented.
☑ All core user stories are documented.
☑ Each core user story has acceptance criteria.
☑ MUST / SHOULD / COULD priorities are agreed.
☑ Core business rules are documented.
☑ Checkout rules are clearly defined.
☑ JWT/current-user rule is clearly defined.
☑ Order ownership/security rules are defined.
☑ Team can start Sprint 2 without unresolved scope questions.
☑ Scope baseline is approved.

Expected Result: SCOPE BASELINE APPROVED